



TWO GUIDES, ONE VOLUME

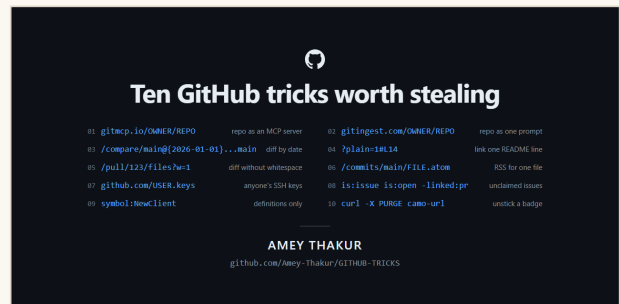
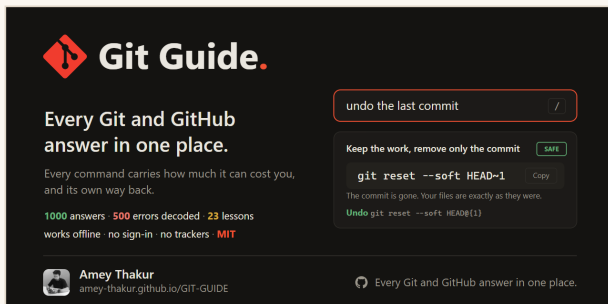
Git and GitHub

Every Git answer, and the GitHub you did not know you had
Free, offline, and dependency-free

1,047
ANSWERS

500
ERRORS DECODED

16
TRICKS SECTIONS



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An independent guide. Not affiliated with the Git project or with GitHub.

A Note from Amey Thakur

Four of the ten highest-voted questions on all of Stack Overflow are Git questions. The top thirty alone have been read about 180 million times. People do not struggle to find Git tutorials; they struggle to find the right command at the moment they need it, and to know whether running it is safe. This guide answers that, for everyone from a student's first push to an engineer rewriting history.

Its companion answers a different question. Where the guide covers what you are trying to do with Git and GitHub, the tricks cover what GitHub can do that you never knew about: the URL parameters, the one-word hostname swaps, the keyboard shortcuts nobody documents and the search syntax with its undocumented limits.

What makes it different

An undo for every command	Where no undo exists, it says so up front, before you run the command rather than after.
Current Git, not 2015 Git	git switch and git restore first, and force-with-lease instead of force.
Real situations, not abstract syntax	Answers split by the case you are in: not pushed, already pushed, teammates have it.
Somewhere to practise	A working Git engine in the browser: branch, merge, rebase, and recover a commit you thought you had destroyed.

Note

Both are free, open source and dependency-free. The site makes no external requests, works offline through a service worker, and sets no trackers.

— Amey Thakur

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Start Here

What is in here.

1,047 ANSWERS	500 ERRORS DECODED	23 LESSONS	16 TRICKS SECTIONS
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Who it is for

You are learning Git	Interactive diagrams, twenty-three lessons and a real Git engine in the page to practise on.
You use Git every day	A thousand answers split by the situation you are actually in, each with its danger level and its undo.
You thought you knew GitHub	Sixteen sections of things it does that are not in the documentation.

💡 Tip

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Four ways in

You need a command now

Type what you want to do, or paste the error Git printed, and copy the answer.

[Open the finder](#)

You are starting from nothing

Zero to your first push in six ordered steps.

[Start](#)

You have broken something

A few questions, then the exact way out.

[Fix](#)

You want to know what GitHub can do

URL tricks, hostname swaps, shortcuts and search syntax.

[GITHUB-TRICKS](#)

How to Use This Guide

Ask, do not browse	The finder takes plain language or a pasted error. It is faster than reading a page.
Read the badge first	Every answer is marked Safe, Rewrites history or Destructive.
Every answer has an undo	Where none exists, it says so before the command, not after.
It works offline	A service worker caches the guide, so it loads with no network and behind corporate proxies.

! Important

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Inside the Guide

Ten pages, each doing one job.

Page	What it does
Finder	Ask anything in plain language, or paste an error. Copy the answer.
Start	Zero to first push in six ordered steps.
Learn	Interactive diagrams: where code lives, what each command moves, branches and HEAD.
Practise	A real Git engine in the page. Twenty-three lessons, a conflict with real markers to delete, and a timed drill that destroys three commits so you can bring them back.
Fix	A few questions, then the exact way out of any mess.
Errors	Git's real error messages, decoded, with fixes.
GitHub	Pull requests, forks, Actions, releases and Pages, with the power moves.
Workflows	How individuals and teams actually run Git, solo to trunk-based.
Cheatsheet	Every command worth knowing, one line each, danger-marked.
Answers	All of them as plain HTML with no JavaScript needed.

Read the Badge Before You Run the Command

Every answer carries one of three danger levels.

Badge	Meaning
Safe	Changes nothing you cannot get back
Rewrites history	Coordinate before doing this to shared branches
Destructive	Can lose work permanently

! Important

Commands marked Destructive can erase work forever. The guide always states the escape route, or the absence of one, before you run the command, never after.

The GitHub You Did Not Know You Had

Its companion repository. Every feature, URL trick, keyboard shortcut, search qualifier and Actions pattern worth knowing, each one verified rather than repeated.

Section	What is in it
Start here	The handful worth knowing before anything else
One-word swaps	Hostname changes that turn a repository into an editor, a prompt, or a server an AI can read
URLs	Parameters and paths that change what a page gives you
Keyboard	The shortcuts, including the ones nobody documents
Search	The syntax, the qualifiers, and the undocumented index limits
Repository	Settings, files and behaviours most repositories never use
Actions	Patterns worth copying, and the ones that will bite
Security	The settings that matter and the ones that only look like they do
Markdown	Everything GitHub's renderer supports, including the parts that are not documented
Profile	What a profile can be made to do
The gh CLI	The commands that replace a browser tab
The API	What is reachable, and what is only reachable there
Git	The parts of Git that GitHub changes
Codespaces	Getting a working machine without one
Copilot	What it can be pointed at
Dead tricks	The ones that used to work, kept so you stop looking

Note

A Dead tricks section records the ones that used to work, so you stop looking for them.

What I Would Tell You

Six things I learned building this, offered for whatever they are worth.

1 Read the danger badge before you run the command.

Every answer carries one of three levels. The one that says Destructive means exactly that, and the guide tells you before you run it, not after.

2 Learn git switch and git restore.

Checkout does four unrelated jobs and that is why it confuses everyone. The newer commands each do one thing.

3 Use force-with-lease, never force.

It refuses to overwrite work that arrived while you were not looking. Plain force does not ask.

4 Practise recovery before you need it.

The drill in the guide destroys three commits and asks you to bring them back. Do it once when it is not real.

5 Learn the reflog.

Almost everything you think you have lost is still there for ninety days. The reflog is how you find it.

6 Read the error, not the internet.

Git's messages are unusually precise once you know the vocabulary. Five hundred of them are decoded in the guide.

💡 Tip

Nobody is born knowing Git. Everyone who looks fluent has simply broken it before and learned the way back.

Explore Further

Where the rest of the work lives.

Related repositories

GIT-GUIDE	The guide this document indexes
Git Guide site	The searchable guide itself
GITHUB-TRICKS	The companion reference
GITHUB-TRICKS list	The whole reference as one page

Elsewhere

GitHub	github.com/Amey-Thakur
Amey's Arc	amey-thakur.github.io
ORCID	0000-0001-5644-1575
Google Scholar	Publications and citations
ResearchGate	researchgate.net/profile/Amey-Thakur
arXiv	arxiv.org/a/thakur_a_3
Kaggle	kaggle.com/ameythakur20
Hugging Face	huggingface.co/ameythakur
LinkedIn	linkedin.com/in/amey-thakur
YouTube	Engineering project demonstrations

A Closing Note

The guide is the work. This is the index to it, written so someone arriving without context can find the part they need.

! **Important**

Git changes, GitHub changes faster, and a trick that works today can be withdrawn tomorrow. The guide is checked and dated; where this document and the guide disagree, the guide is right.

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